

# Attack-Family-Dependent Zero-Day Generalization in Network Intrusion Detection: A Comparative Analysis of Classical and Deep Learning Models

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**Abstract** - Network Intrusion Detection Systems (NIDS) based on deep learning have proven highly efficient in classifying known network attacks. Nevertheless, the generalization abilities of the models to unseen attack families appear to be problematic. This research paper explores attack-family-dependent zero-day generalization capabilities of the CNN-GRU-based NIDS on the CICIDS2017 dataset using ANOVA-based feature selection and the Top-30 feature configuration for evaluation. In this paper, we use the Leave-One-Attack-Out (LOAO) evaluation approach where six different attack families were excluded from training data one by one and evaluated as unseen classes. Results show a significant discrepancy between conventional closed-set classification and unseen attack families generalization. In the context of six attack families under investigation, the mean recall was found to be equal to 6.23%, and the mean F1-score was equal to 11.50%. Moreover, the generalization to different attack families varies significantly, proving that generalization to unseen attack families is not homogeneous. For a more detailed qualitative analysis, we applied t-SNE to the Top-30 feature space.

**Keywords** - Network Intrusion Detection, Deep Learning, Zero-Day Attack, CNN, GRU, CNN-GRU, Attack Generalization, CICIDS2017

The proliferation of interconnected computer networks and Internet-based services has increased the vulnerability of organizational and communication infrastructures to cyber attacks. Malicious activities like denial of service attacks, port scanning, brute force attacks, botnet activities and web based attacks can compromise availability, confidentiality and integrity of network resources. Hence, Network Intrusion Detection Systems (NIDS) are an indispensable part of cybersecurity infrastructure, since they examine network traffic and recognize patterns related to malicious activities.

Existing intrusion detection methods rely on pre-defined signatures or manually constructed rules. These methods can identify known attacks successfully but have limited capabilities to detect unknown or evolving attack patterns. Therefore, machine learning (ML) and deep learning (DL) techniques have been widely studied for network intrusion detection because of their ability to learn patterns from network traffic and automatically detect malicious activities [2] [8] [1]. Traditional classifiers and deep learning models, such as Convolutional Neural Networks (CNNs), Gated Recurrent Units (GRUs), Long Short-Term Memory (LSTM), and hybrid architectures, have been applied in recent studies for intrusion detection [1], [2].

Many learning-based NIDS approaches report high classification performance, but their performance can drop when they encounter attack types not present

## I. INTRODUCTION

during training. This problem is especially important in the case of zero-day or unseen attacks, where the characteristics of an attack may be different from the patterns learned from known attack families [3] [5]. Therefore, good performance on traditional random train-test splits does not necessarily imply reliable detection of new attack families. This leads to an inconsistency between the performance on the traditional benchmark and the generalization ability required in practical intrusion detection. Prior research has examined zero-day intrusion detection with different evaluation strategies, including removing some types of attacks from the training data and subsequently testing the models on these unseen attack types [4]. For example, existing comparative studies [3], [4] have shown that treating attack categories as unseen during evaluation leads to substantial degradation in detection performance. However, most of this work is primarily concerned with either measuring the resulting performance drop or comparing detection models, rather than exploring the reasons for the variation in generalization performance across different attack families. This motivates a more detailed analysis of the features associated with successful or unsuccessful zero-day generalization.

To study this issue, this study uses the CICIDS2017 dataset that is widely used to evaluate network intrusion detection methodologies [3], which contains labeled benign traffic and different current attack categories. Its attack families diversity makes it an appropriate dataset to test the generalization of models trained on known attack patterns to an excluded attack family. The evaluation strategy employed in this study is based on the Leave-One-Attack-Family-Out (LOAO) approach. In this protocol, one attack family is completely excluded from the training process and is treated as an unseen attack during testing. We repeat this process across several sufficiently represented attack families, which makes it possible to study if the zero-day generalization behavior is consistent across different attack types. The study compares a classical machine learning baseline, Random Forest (RF) to three deep learning architectures: CNN, GRU and hybrid CNN-GRU. We consider three DL architectures to compare convolutional, recurrent and hybrid representations under the same evaluation protocol with unseen attacks, and include Random Forest as a non-deep-learning baseline. The goal is not to propose a new CNN-GRU architecture, but rather to see if different model families

display similar or different patterns of attack-family-dependent generalization.

In addition to traditional performance evaluations, this work explores the features related to the performance of zero-day detection. Specifically, feature distribution is examined to determine whether the unseen attack traffic shares similarities with the benign traffic or seen attack traffic. Additionally, dimensionality reduction of the learned representations by deep learning models will be performed using t-SNE and UMAP. This will give us insight into the relationship between the feature space and representation space of the seen and unseen attack families.

The main contributions of this study are as follows:

- An attack-family-dependent zero-day evaluation of NIDS models using a repeatable Leave-One-Attack-Family-Out (LOAO) protocol across multiple attack families.
- A comparative evaluation of Random Forest, CNN, GRU, and CNN-GRU under identical unseen-attack evaluation conditions.
- An analysis of raw network features and learned representations to investigate factors associated with successful or unsuccessful zero-day generalization.
- An empirical characterization of attack families that are difficult to generalize to, providing insights that can guide the development of more robust deep learning-based NIDS approaches.

The rest of this paper will be structured in the following way. The second section will give a survey of the previous research done on machine learning and deep learning based network intrusion detection systems, especially in regard to the detection of unknown attacks and zero-day attacks. The methodology of the research and LOAO experiment will be presented in Section III. The experimental setup will be described in Section IV. The results of the experiment will be shown in Section V. The analysis of the generalization properties will be carried out in Section VI.

## II. RELATED WORK

### A. Intrusion Detection using Deep Learning

Various studies have been conducted on hybrid deep learning architectures for intrusion detection. A hybrid CNN-LSTM framework was proposed for cyber intrusion identification by integrating convolutional

feature extraction and recurrent sequence modeling [1]. Similarly, sequential deep learning approaches have been explored to improve the robustness of NIDS for different network attack types [2]. Classical machine learning and deep representation-based models have also been compared on benchmark datasets such as NSL-KDD and CICIDS2017, highlighting the fact that model performance may vary greatly across datasets and attack categories [3]. Other studies have looked at different deep learning architectures and feature-selection strategies. Transformer-based methods have been studied to capture relationships among network features, and feature-selection methods have been applied to reduce dimensions and computational load [5], [7]. The systematic review of deep learning-based NIDS shows that CNNs, recurrent networks, autoencoders, and hybrid architectures have become popular methods for network intrusion detection [6]. These works show the success of deep learning for intrusion detection, but most evaluations use the conventional train-test setup, where the same attack categories are available in both the training and testing datasets. Such assessment does not necessarily reflect the difficulty of detecting a previously unseen attack family.

### ***B. Zero-Day and Unseen Attack Detection***

A critical need for a practical NIDS is the ability to detect previously unseen attacks, as new attacks may not be available in the training data. Some studies have therefore tried to simulate zero-day scenarios by removing some attack categories from training and then testing the trained model on those unseen categories. A study for feature selection considered a CNN-LSTM-DQN intrusion detection framework for zero-day and drifted network traffic and showed that the detection performance can vary significantly with a change in the traffic distribution [4]. Similarly, the comparative studies of intrusion detection models have shown that models that have good performance under traditional evaluation can degrade significantly when attack categories are considered unseen during testing [3]. Methods of feature-selection and representation learning have been explored as well to improve the intrusion detection performance under changing traffic conditions [5]. Common feature selection and deep autoencoders-based approaches have been optimized to reduce the feature space while keeping discriminative information for IoT intrusion detection. Other work has studied wrapper-based feature selection with attention-based transformer models to enhance the representation

of network traffic [7]. An important observation of these studies is that zero-day performance does not necessarily generalize across attack categories. In particular, the model's ability to generalize to an unseen attack can be affected by the characteristics of that attack family and its similarity to the traffic patterns seen during training. However, previous work has mainly focused on reporting detection performance or comparing different detection architectures, and comparatively less work has been devoted to systematically studying why generalization performance varies across unseen attack families.

### ***C. Feature Selection and Representation Analysis***

Feature selection is an important step in NIDS because network traffic datasets usually have a large number of features, some of which may be redundant or weakly informative. Reducing the feature space can reduce the computational needs and can improve the ability of a model to learn meaningful traffic patterns. Therefore, statistical, wrapper-based, and model-based feature selection approaches have been studied in previous works for intrusion detection [5], [7]. In addition to selecting informative input features, deep learning models also convert the original input into internal representations through the hidden layers. Such learned representations can capture information about how different traffic categories are separated in the feature space of the model. Most NIDS studies, however, assess these models mainly on classification metrics like accuracy, precision, recall, and F1-score. The performance indicators themselves do not explain why a particular attack family is difficult to detect after it is removed during training. Analyzing the distribution of raw features allows us to determine whether an unseen attack is similar to any known attacks or benign traffic. Similarly, visualization techniques such as t-SNE and UMAP can help determine whether the learned representations form distinct clusters or overlap between attack families. Thus, such analysis can provide further insight into the relationship between feature characteristics and zero-day generalization.

### ***D. Research Gap***

The studies we have examined indicate that the zero-day detection performance can vary significantly when attack categories are excluded from the model training [3], [4]. However, the reasons behind this variation over different unseen attack families are still not well explored. There is also limited controlled comparison between classical and deep learning models

under the same attack-family exclusion protocol in existing research. Additionally, classification metrics do not inform us whether poor generalization is due to statistical properties of a previously unseen attack or how the attack is represented in the model's learned feature space. To address these limitations, we perform an attack-family-level analysis of zero-day generalization on the CICIDS2017 dataset. Random Forest is used as a classical machine learning baseline. CNN, GRU, and CNN-GRU are evaluated as deep learning models under the Leave-One-Attack-Family-Out (LOAO) protocol. Raw feature distributions and learned representations are analyzed using statistical feature analysis and dimensionality-reduction techniques such as t-SNE and UMAP in addition to traditional metrics for performance.

### III. DATASET AND EXPERIMENTAL SETUP

#### A. Dataset Description

The experiments were performed on the CICIDS2017 dataset which is a labeled network intrusion detection dataset that contains benign traffic and different types of attacks. Seven traffic classes with 61 usable features were considered after data cleaning and pre-processing. The reason we selected this dataset is that it covers a broad range of attack categories, allowing us to control the evaluation of generalization to unseen attack families.

TABLE I

Summary of the CICIDS2017 Dataset Used in the Present Study

| Class         | Traffic Type   | Used in LOAO |
|---------------|----------------|--------------|
| BENIGN        | Normal traffic | No           |
| DoS Hulk      | Attack         | Yes          |
| DDoS          | Attack         | Yes          |
| PortScan      | Attack         | Yes          |
| DoS GoldenEye | Attack         | Yes          |
| FTP-Patator   | Attack         | Yes          |
| SSH-Patator   | Attack         | Yes          |

#### B. Data Preprocessing

The dataset was cleaned by handling invalid and missing values and preparing the traffic records for the model training. Feature scaling was done using StandardScaler and was fitted on training data only to prevent data leakage. The class labels were encoded to be used for multi-class classification.

#### C. Feature Selection

ANOVA F-test was used to find features with stronger class discrimination. Experiments were conducted with different numbers of selected features, e.g., 10, 20, 30, 40 and all available features. Based on the experimental results, the 30-features configuration was chosen for the detailed analysis.

#### D. Sequence Construction and Model Input

To encode the temporal relationships of network traffic, the features used for the CNN-GRU model were structured in fixed-length sequences. Sequence length was 10.

#### E. Leave-One-Attack-Out Evaluation

We used a Leave-One-Attack-Out (LOAO) protocol to simulate the zero-day detection. In each experiment, one attack family was completely left out from the training data and was used only for evaluation. Using this protocol, six attack families were evaluated: DoS Hulk, DDoS, PortScan, DoS GoldenEye, FTP-Patator and SSH-Patator.

#### F. Evaluation Metrics

The model was evaluated in terms of accuracy, precision, recall, F1-score and false positive rate. We paid special attention to recall and F1-score for the held-out attack family, as they show how well the model can recognize attacks that it has never seen before.

### IV. MODELS AND METHODOLOGY

#### A. Overall Methodology

The proposed experimental framework assesses the generalization ability of deep learning models on unseen attack families. The workflow includes preprocessing of data, feature selection through ANOVA, construction of sequences, training of CNN-GRU model and Leave-One-Attack-Out (LOAO) evaluation. In each LOAO experiment, one attack family is excluded from the training process and serves as an unseen class during the test.

#### B. CNN-GRU Model

Hybrid CNN-GRU architecture for intrusion classification. The CNN part extracts local patterns from the selected network-flow features, while the GRU part captures sequential relationships in the constructed traffic sequences. The extracted features are passed from the CNN layer to the GRU layer to learn the temporal information present in the input sequence. The resulting representation is then fed to a fully connected classification layer to predict the traffic class.

### C. Training Procedure

The prepared traffic sequences were used to train the CNN-GRU model for multi-class classification. The training data was kept separate from the data used for evaluation. Feature-scaling parameters were obtained from the training data only and then applied to the corresponding evaluation data to prevent data leakage. The model was trained using the attack categories available in the training set and was subsequently used to classify the evaluation samples. The same model configuration was maintained across the experiments to ensure consistent evaluation.

### D. Leave-One-Attack-Out Testing

The zero-day evaluation is performed via the LOAO protocol. In each experiment, we remove one whole attack family from the training set and train the model with the rest of the attack families. The excluded attack family is then tested as an unseen class. This procedure is used to independently evaluate six attack families, namely DoS Hulk, DDoS, PortScan, DoS GoldenEye, FTP-Patator, and SSH-Patator. This setup allows the study to analyze how the zero-day generalization performance varies across attack families and to identify attack classes that are particularly hard for the model to recognize when unseen during training.

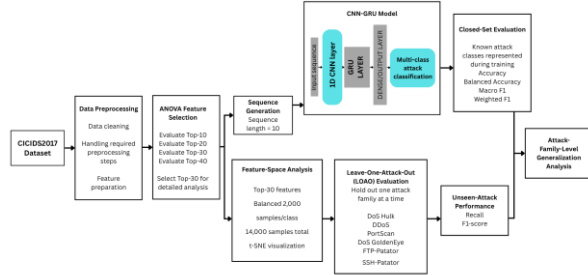


Fig. 1. Overall methodology of the CNN-GRU-based attack-family-level zero-day generalization analysis.

## V. EXPERIMENTAL PROTOCOL

### A. Closed-Set Evaluation

Used the CNN-GRU model to test different feature configurations selected by ANOVA: Top-10, Top-20, Top-30, Top-40 and all 61 features available. The classification performance was assessed using accuracy, balanced accuracy, macro F1-score, and weighted F1-score.

### B. Unseen-Attack Evaluation

In the unseen-attack experiments, six attack families were tested individually: DoS Hulk, DDoS, PortScan,

DoS GoldenEye, FTP-Patator and SSH-Patator. The chosen attack family was treated as an unseen class and was not included in the training of the model for the corresponding experiment and was evaluated separately. The attention was paid to recall and F1-score as the measures of model's ability to recognize the excluded attack family.

### C. Feature-Space Analysis

For the feature space analysis the Top-30 feature configuration was used. For each of the 7 classes, a balanced subset of 2,000 samples was used, for a total of 14,000 samples. The selected feature space was represented in two dimensions using t-SNE to visually inspect the distribution and overlap of the traffic classes.

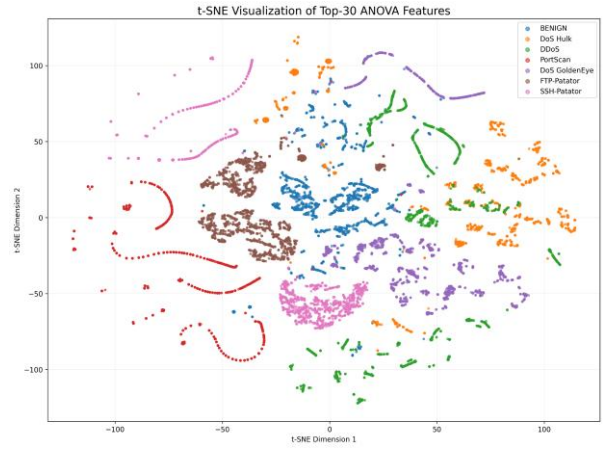


Fig. 2. t-SNE visualization of the Top-30 feature representation across traffic classes.

### D. Experimental Audit

Additional investigations were carried out to study the preparation of the experimental data. The audit found significant overlap between training and test feature rows, including 45,848 common Top-30 feature rows and 81 label-inconsistent duplicate signatures. These findings are taken into account when interpreting the traditional evaluation results, especially the very high closed-set performance.

## VI. RESULTS AND ANALYSIS

### A. Feature Selection Results

The CNN-GRU model was evaluated with different ANOVA based feature configurations like Top-10, Top-20, Top-30, Top-40 and all 61 features. The results showed that the reduced feature configurations preserved strong classification performance in the conventional evaluation. Therefore, we chose the Top-

30 setting for the subsequent detailed study and for the unseen-attack experiments.

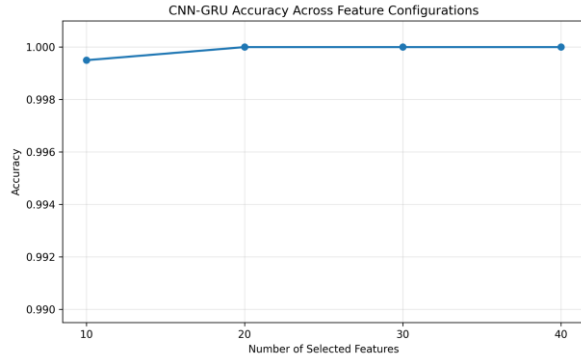


Fig. 3. Classification accuracy for different feature configurations.

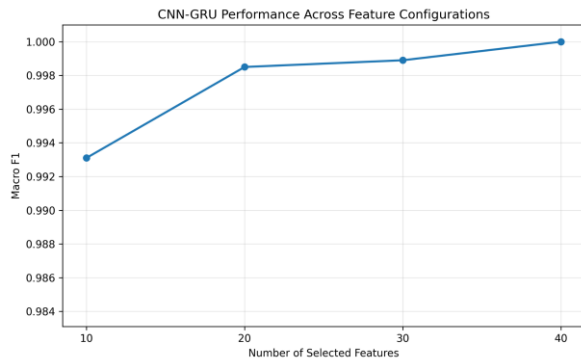


Fig. 4. Macro F1-score for different feature configurations.

The comparison shows the effect of reducing the original feature set on model performance. Based on the overall feature-selection evaluation, the Top-30 configuration was selected for the subsequent detailed analysis and unseen-attack experiments.

### B. Closed-Set Classification Results

In the conventional evaluation setting, the CNN-GRU model achieved very high classification performance with accuracy and F1-score approaching 100%. These results indicate that the model can correctly classify attack families when the corresponding classes exist during training. The experimental audit, however, found a substantial overlap of feature rows between training and testing data. Hence, these results should be considered as closed-set performance rather than evidence for strong generalization to truly unseen traffic.

### C. Unseen-Attack Generalization Results

The LOAO experiments yielded a very different result than the closed-set evaluation. On the six held-out attack families, the CNN-GRU achieved a mean recall of 6.23% and a mean F1-score of 11.50%. This

large decrease suggests that high conventional classification performance does not guarantee that the model can effectively recognize attack families unseen during training.

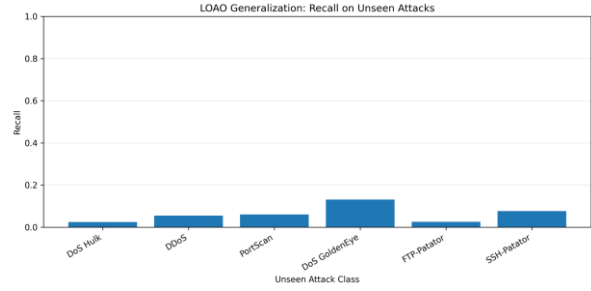


Fig. 5. Recall of the CNN-GRU model across held-out attack families.

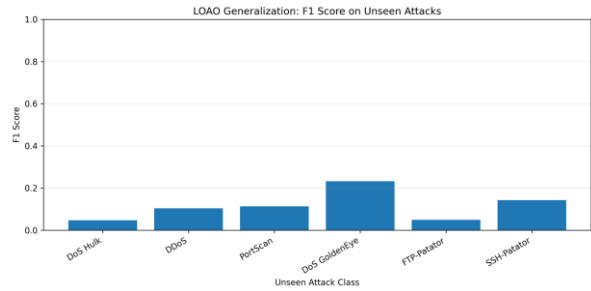


Fig. 6. F1-score of the CNN-GRU model across held-out attack families.

### D. Attack-Family-Level Performance

Results differed significantly across the held-out attack families. DoS Hulk, DDoS and PortScan had relatively higher recall than the other held-out families. DoS GoldenEye, FTP-Patator and SSH-Patator demonstrated considerably weaker generalization. This variation suggests that zero-day performance is not uniform across attack families and motivates the study of the characteristics of the corresponding feature representations.

### E. Feature-Space Analysis

Further investigated the Top-30 feature configuration on a balanced subset of 14,000 samples with t-SNE. The visualization was used for a qualitative analysis of the separation and overlap among traffic classes. This provides a feature-space insight to understand why some attack families might be more difficult to recognize when they are not part of the training. The t-SNE visualization however is used as a qualitative diagnostic and not a quantitative proof for class separability.

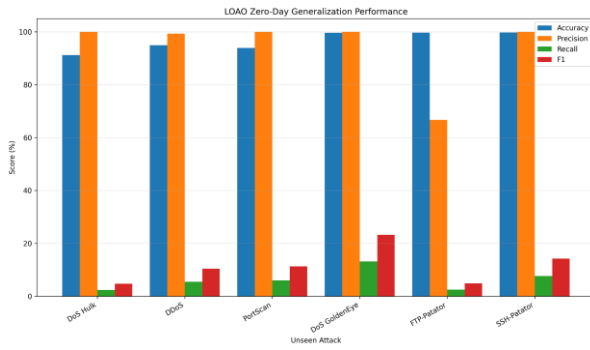


Fig. 7. Comparative LOAO performance across the evaluated attack families.

## VII. DISCUSSION

As can be seen in the experimental results, there is a significant gap between regular classification and zero-day generalization to unseen attack families. While the CNN-GRU model obtained extremely good performance in the case where the attack families used during testing were present during training, its performance has been greatly decreased when some attack families were missing during training. The mean recall of 6.23% and mean F1-score of 11.50% show that high performance in a closed set of attacks does not guarantee zero-day generalization. Moreover, the performance of the model was highly variable across the tested attack families. DoS Hulk, DDoS, and PortScan were relatively well generalizable, while DoS GoldenEye, FTP-Patator, and SSH-Patator have poor performance. It shows that recognition of an unseen attack is dependent on which exact attack family it belongs to and not on the fact that the attack is unseen.

The t-SNE visualization helps to analyze these differences qualitatively using the feature space view. Thus, it is possible to find out if the hard-to-detect attack families occupy a different part of the feature space, which is less discriminative in terms of patterns encountered during training. Still, t-SNE is used solely for the analysis of the phenomenon and cannot reveal the reason behind it. It should be noted that the evaluation of the intrusion detection models is performed using the attack family level of protocol for unseen attack evaluation, as it allows obtaining more accurate results. Regular train-test protocol can give an excessively optimistic view of model performance, while exclusion of attack families is a more difficult test of its generalization capabilities. The experimental audit also provides an important consideration when interpreting the closed-set results. The presence of overlapping feature rows between the training and test

data indicates that some traffic records share identical feature representations across the two sets, while the label-inconsistent duplicate signatures show that identical feature representations can be associated with different labels. Therefore, the very high closed-set performance should be interpreted with caution, as it may not fully reflect the model's ability to generalize to genuinely unseen traffic. In contrast, the LOAO evaluation provides a stricter assessment by completely excluding the selected attack family from the training process.

In summary, this study highlights the significance of zero-day generalization analysis based on attack families and feature space investigation.

## VIII. CONCLUSION AND FUTURE WORK

In this paper, we studied whether the CNN-GRU-based network intrusion detection model can generalize to unseen attack classes. We demonstrated through experiments using CICIDS2017 dataset and LOAO evaluation scheme a significant gap between the conventional closed-set classification and the performance on unseen attacks. While the performance of the model on attack classes was extremely high for those that were used during training, the model showed the mean recall and F1-score of 6.23% and 11.50%, correspondingly, for the six held-out attack families.

Moreover, we demonstrated that the zero-day generalization varies significantly among attack families. This is important since it means that one cannot simply evaluate intrusion detection models using a conventional random train-test evaluation. Additionally, feature space analysis was performed using t-SNE to identify the differences in representation of traffic classes and gain qualitative insight into generalization behavior of the model.

For future work, we plan to further improve the generalization of deep learning-based intrusion detection models to unseen and evolving attacks.

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